PROBLEM CL19

Elbow Mixing Two Air Streams: Outlet Velocity, Force and Heat

Original statement · CL19.pdf

Coflow | Perfect gas | Linear momentum | Heat

OFFICIAL PROBLEM STATEMENT

The elbow-shaped tube of circular cross section shown in the figure of CL19.pdf serves to mix two different air streams. The pipe discharges to the open atmosphere after mixing is completed. At the inlet, where the pressure $p_i = 2p_a$ and velocity U_i are uniform, the inner stream, with diameter $D/\sqrt{2}$, has temperature T_1, and the outer coflow stream has temperature T_2. The temperature measurement at the outlet reveals that $T_o = T_1T_2/(T_1 + T_2)$.

- Determine the uniform gas velocity at the outlet U_o.
- Compute the force $\bar{F} = F_x\bar{e}_x + F_y\bar{e}_y$ acting on the pipe. In the computation, neglect the effect of gravity.
- Obtain the amount of heat transferred per unit time through the pipe wall.

Note: for a perfect gas, $p = \rho R_g T$, where R_g is the gas constant.

Source: CL19.pdf (page 1, prefinal of Dec 16, 2013). Notes.pdf, Chapter 3: integral mass balance Eq. 3.11, integral momentum balance Eq. 3.36, integral total energy balance Eq. 3.46.

Worked solution (4-phase method)

PHASE 1

Hypotheses, Data and Unknowns

- Steady; perfect gas, $\rho = p/(R_gT)$, $h = c_pT$; neglect $\bar{g}$; uniform velocity and pressure at the inlet; uniform state at the outlet (mixing completed); no moving parts.
- Inlet (diameter D, flow along $+\bar{e}_x$): inner stream of area $\pi D^2/8$ at T_1, outer annulus of area $\pi D^2/4 - \pi D^2/8 = \pi D^2/8$ at T_2, both at U_i and $p_i = 2p_a$. Outlet (diameter D, flow along $+\bar{e}_y$): free jet at p_a, T_o.
- Data: $[p_a,\ U_i,\ T_1,\ T_2,\ D,\ R_g,\ c_p]$. Unknowns: $[U_o,\ F_x,\ F_y,\ \dot{Q}]$.

Symbol	Meaning	SI unit
U_i, U_o	inlet and outlet velocities	m/s
T_1, T_2, T_o	temperatures	K
D	pipe diameter	m
$\dot{Q}$	heat from the wall to the gas	W

PHASE 2

Frames, Control Volume and Surfaces

- Frame fixed to the pipe, $\bar{e}_x$ horizontal (inlet flow), $\bar{e}_y$ vertical upwards (outlet flow). Fixed control volume. $A \equiv \pi D^2/4$.
- Surfaces: Σ_i (inlet, $\bar{n} = -\bar{e}_x$), Σ_o (outlet, $\bar{n} = +\bar{e}_y$), Σ_ℓ (pipe wall).

$$\boxed{\bar{F} = \int_{\Sigma_\ell} \left[(p - p_a)\, \bar{n} - \bar{\bar{\tau}}' \cdot \bar{n} \right] d\sigma}$$

$$\boxed{\dot{Q} = -\int_{\Sigma_\ell} \bar{q} \cdot \bar{n}\, d\sigma}$$

PHASE 3

Balances, Surface by Surface

1. Mass (Notes Eq. 3.11)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c} \rho\, dV}}_{0 \text{ (steady)}} + \int_{\Sigma_c} \rho \left(\bar{v} - \underbrace{\cancel{\bar{v}_c}}_{0} \right) \cdot \bar{n}\, d\sigma = 0$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\,\bar{v}\cdot\bar{n}\,d\sigma$
Σ_i (T_1)	$-\frac{p_i}{R_g T_1} U_i \frac{\pi D^2}{8}$
Σ_i (T_2)	$-\frac{p_i}{R_g T_2} U_i \frac{\pi D^2}{8}$
Σ_o	$+\frac{p_a}{R_g T_o} U_o \frac{\pi D^2}{4}$
Σ_ℓ	0

With $p_i = 2p_a$: $\frac{p_a}{R_g} U_i \frac{\pi D^2}{4}\left(\frac{1}{T_1}+\frac{1}{T_2}\right) = \frac{p_a}{R_g T_o} U_o \frac{\pi D^2}{4}$, and $\frac{1}{T_1}+\frac{1}{T_2} = \frac{T_1+T_2}{T_1 T_2} = \frac{1}{T_o}$.

$$\boxed{U_o = U_i} \tag{1}$$

2. Linear momentum (Notes Eq. 3.36)

$$\underbrace{\frac{d}{dt}\cancel{\int_{V_c}\rho\bar{v}\,dV}}_{0\ \text{(steady)}} + \int_{\Sigma_c}\rho\bar{v}\,[(\bar{v}-\underbrace{\cancel{\bar{v}_c}}_{0})\cdot\bar{n}]\,d\sigma = -\int_{\Sigma_c}(p-p_a)\,\bar{n}\,d\sigma + \int_{\Sigma_c}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma + \underbrace{\cancel{\int_{V_c}\rho\bar{g}\,dV}}_{\text{neglect }\bar{g}}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\bar{v}\,(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int (p-p_a)\,\bar{n}\,d\sigma$	$\int \bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$
Σ_i	$-\frac{p_i}{R_g}\left(\frac{1}{T_1}+\frac{1}{T_2}\right) U_i^2 \frac{\pi D^2}{8}\bar{e}_x = -\frac{p_a}{R_g T_o} U_i^2 A\,\bar{e}_x$	$(p_i - p_a) A\,\bar{e}_x = p_a A\,\bar{e}_x$	0
Σ_o	$+\frac{p_a}{R_g T_o} U_o^2 A\,\bar{e}_y$	0 (free jet)	0
Σ_ℓ	0	$-\int_{\Sigma_\ell}(p-p_a)\,\bar{n}\,d\sigma$	$\int_{\Sigma_\ell}\bar{\bar{\tau}}'\cdot\bar{n}\,d\sigma$

Boxed Σ_ℓ terms: $-\bar{F}$. With (1), $U_o = U_i$.

$$\boxed{\bar{F} = \frac{\pi D^2}{4}\,p_a\left(1+\frac{U_i^2}{R_g T_o}\right)\bar{e}_x - \frac{\pi D^2}{4}\,\frac{p_a U_i^2}{R_g T_o}\,\bar{e}_y} \tag{2}$$

Erratum in the official solution (sign of F_y)

CL19.pdf writes the momentum balance with $+\frac{p_a}{R_g T_o} U_o U_o \bar{e}_y \frac{\pi D^2}{4}$ on the left-hand side and $-\bar{F}$ on the right, but boxes $\bar{F}$ with $+\frac{\pi D^2}{4} p_a \frac{U_i^2}{R_g T_o} \bar{e}_y$. Moving the outlet flux to the other side gives the minus sign of (2): the jet leaves upwards, so its reaction on the pipe points downwards. The computed sign is kept.

3. Total energy (Notes Eq. 3.46)

$$\underbrace{\frac{d}{dt}\int_{V_c} \rho\left(e+\tfrac{v^2}{2}\right) dV}_{0\ \text{(steady)}} + \int_{\Sigma_c} \rho\left(e+\frac{v^2}{2}\right)\left[(\bar{v}-\underbrace{\bar{v}_c}_{0})\cdot\bar{n}\right] d\sigma = -\int_{\Sigma_c} p\,\bar{v}\cdot\bar{n}\,d\sigma + \int_{\Sigma_c} \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma + \underbrace{\int_{V_c} \rho\bar{g}\cdot\bar{v}\,dV}_{\text{neglect } \bar{g}} - \int_{\Sigma_c} \bar{q}\cdot\bar{n}\,d\sigma + \underbrace{\int_{V_c} Q_R\,dV}_{0}$$

Each surface integral, surface by surface (the vertical line drawn under each integral in class):

Surface	$\int \rho\left(e+\frac{v^2}{2}\right)(\bar{v}\cdot\bar{n})\,d\sigma$	$-\int p\,\bar{v}\cdot\bar{n}\,d\sigma$	$\int \bar{v}\cdot(\bar{\bar{\tau}}'\cdot\bar{n})\,d\sigma$	$-\int \bar{q}\cdot\bar{n}\,d\sigma$
Σ_i (T_1)	$-\dot{m}_1\left(e_1+\frac{U_i^2}{2}\right)$	$+p_i U_i \frac{\pi D^2}{8}$	0	0
Σ_i (T_2)	$-\dot{m}_2\left(e_2+\frac{U_i^2}{2}\right)$	$+p_i U_i \frac{\pi D^2}{8}$	0	0
Σ_o	$+\dot{m}\left(e_o+\frac{U_o^2}{2}\right)$	$-p_a U_o A$	0	0
Σ_ℓ	0	0	0	$-\int_{\Sigma_\ell} \bar{q}\cdot\bar{n}\,d\sigma$

Boxed term: $\dot{Q}$. Combining with the pressure work, $h = c_p T$; the kinetic terms cancel ($U_o = U_i, \dot{m}_1 + \dot{m}_2 = \dot{m}$). Enthalpy fluxes: $\dot{m}_k c_p T_k = c_p \frac{p_i}{R_g} U_i \frac{\pi D^2}{8}$ for each inlet stream, so their sum is $c_p \frac{2p_a}{R_g} U_i A$; at the outlet $\dot{m} c_p T_o = c_p \frac{p_a}{R_g} U_i A$.

$$\dot{Q} = c_p \frac{p_a}{R_g} U_i A - c_p \frac{2p_a}{R_g} U_i A \implies \boxed{\dot{Q} = -\frac{c_p}{R_g}\, p_a U_i\, \frac{\pi D^2}{4}} \qquad (3)$$

PHASE 4

Interpretation, Limits and Dimensional Check

- $\dot{Q} < 0$: heat leaves the gas through the wall. This is consistent with the data: $T_o = T_1 T_2/(T_1+T_2)$ is lower than both inlet temperatures, so the gas must be cooled (the official solution notes that the fluid is cooled down as it flows through the pipe).

- The outlet velocity equals the inlet one because the density halves with the pressure (p_a vs $2p_a$) while the temperature effect is exactly offset by the chosen T_o.
- Dimensions (SI): $[p_a A] = \mathrm{N}$; $[p_a U_i^2 A/(R_g T)] = \mathrm{N}$; $[c_p p_a U_i A/R_g] = \mathrm{W}$.
- Numerical illustration (hypothetical data): $p_a = 101325$ Pa, $U_i = 20$ m/s, $T_1 = 300$ K, $T_2 = 400$ K ($T_o = 171.43$ K), $D = 0.1$ m, $R_g = 287$ J/(kg K), $c_p = 1005$ J/(kg K) give $F_x = 802.27$ N, $F_y = -6.47$ N and $\dot{Q} = -55734$ W.